

Fusion-Integrated · Module Structure

Placement Cell as the reference. Every other module keeps this shape — the boundaries below are enforced by CI, not by convention.

The five layers

domain/	Pure Python. Entities and rules. No Django, so it tests in milliseconds with no database.
models.py	ORM only. No business logic. No ForeignKey leaving the module.
selectors/ not a 403.	All reads. Authorisation happens here by narrowing the queryset, which is why an out-of-scope row is a 404 and not a 403.
services/	All writes. Transaction boundaries. Emits notifications and events.
api/	Thin: parse, scope, delegate, serialise.

Why the split matters. Read scope and write authority are different questions. A student can legitimately see every published posting, so passing that queryset into a service proves only that the row is reachable — never that the caller may change it. Services therefore call `authz.require(actor, <permission>)` as well.

Contracts CI enforces

RULE	CHECKED BY
A module may import another ONLY via <code>modules.<other>.contracts</code>	<code>import-linter</code> , 4 contracts
No ForeignKey may cross a module boundary — use a plain integer id	<code>ops/checks/no_cross_module_fk.py</code>
<code>contracts.py</code> getters are plural: <code>get_users(ids)</code> , never <code>get_user(id)</code>	<code>ops/checks/contracts_are_plural.py</code>
<code>domain/</code> must not import Django	<code>import-linter</code>
Every server nav item needs a client route, and vice versa	<code>ops/checks/nav_matches_routes.py</code>
The committed OpenAPI schema must match the code	<code>make schema-check</code>

Generated from the real tree by `make module-structure`. Regenerate it rather than editing the PDF — a hand-kept copy is wrong the first time someone adds a service.

Backend — modules/placement/

└─ api/	THIN. Parse, scope, delegate, serialise. No business logic.
└─ academics.py	The CPI directory. Staff only — the largest disclosure here.
└─ audit.py	The trail. Staff see reasons and actors; a student sees the timeline.
└─ clearance.py	A student sees only their own clearance — no user parameter exists.
└─ conduct.py	Recording an incident and imposing a sanction are separate.
└─ documents.py	Attach a Drive link; reaching one re-checks scope.
└─ exports.py	CSV. Escaping lives in core.api.csv so no export forgets it.
└─ permissions.py	Module-specific permission classes.
└─ registration.py	Self-service registration; fee routes are staff-only.
└─ serializers.py	Explicit allow-lists. `fields = "__all__"` is never used.
└─ urls.py	Mounted by config/urls.py under /api/v1/.
└─ views.py	Endpoints. Each takes its queryset from selectors/.
└─ domain/	PURE PYTHON. No Django import — CI fails the build on one.
└─ clearance.py	Rules 22 and 24: the no-dues gate and non-joining.
└─ conduct.py	The debarment ladder (rules 18/19/21).
└─ eligibility.py	The rule AST, evaluated fail-closed.
└─ offer_policy.py	The signed policy as rules. Testable in milliseconds.
└─ profile_completeness.py	Weights and the named list of what is missing.
└─ registration.py	Who may register, and on what terms.
└─ state_machine.py	Legal transitions as a table; an illegal one cannot be expressed.
└─ management/	Operational commands, not business logic.
└─ commands/	
└─ send_notifications.py	Drain or inspect the outbox by hand.
└─ migrations/	One per schema change. Never edited after it ships. 8 so far.
└─ selectors/	ALL reads. Every queryset a view uses is built here.
└─ applications.py	Read helpers for the application list.
└─ scoping.py	Who may see what. Narrowing means a foreign row is 404, not 403.
└─ services/	ALL writes. Transaction boundaries live here.
└─ announcements.py	Publish and withdraw; history is kept.
└─ applications.py	The only caller of the state machine.
└─ authz.py	Write authority. `scope` proves a row is readable, not writable.
└─ clearance.py	Rule 24's no-dues lever, exposed to other modules via contracts.py.
└─ companies.py	Registration and the PC-BR-007 approval gate.
└─ conduct.py	Records incidents; imposing a sanction is a separate act.
└─ documents.py	Attach, replace, remove a document link.
└─ facts.py	Assembles eligibility facts in a fixed query count.
└─ interviews.py	Rounds, candidates and outcomes.
└─ notifications.py	Outbox writes and the drain.
└─ offers.py	Accept/issue, with the per-student mutex.
└─ postings.py	Create, update, publish. Criteria freeze on publish.
└─ profiles.py	Profile and derived completeness.
└─ recruiters.py	Invitation, acceptance, sign-in.
└─ registration.py	Rules 1, 20, 21.
└─ stats.py	Snapshots, with small-cell suppression.
└─ tests/	domain/ 90%, services 85%, api 80%.
└─ test_api.py	Tests: api.
└─ test_audit.py	Tests: audit.
└─ test_bulk_actions.py	Tests: bulk actions.
└─ test_clearance.py	Tests: clearance.
└─ test_conduct.py	Tests: conduct.
└─ test_cpi_directory.py	Tests: cpi directory.
└─ test_documents.py	Tests: documents.
└─ test_domain.py	Pure, no database, milliseconds.
└─ test_eligibility_rules.py	Tests: eligibility rules.
└─ test_exports.py	Tests: exports.
└─ test_notification_worker.py	Tests: notification worker.
└─ test_offer_flow.py	Tests: offer flow.
└─ test_offer_policy.py	Tests: offer policy.
└─ test_privilege_escalation.py	Read scope is not write authority.
└─ test_recruiter_isolation.py	Tests: recruiter isolation.
└─ test_registration.py	Tests: registration.
└─ test_schedule.py	Tests: schedule.
└─ apps.py	AppConfig. Registers this module's system checks.
└─ authentication.py	Only if the module owns a credential pool. Most do not.
└─ contracts.py	THE ONLY module-to-module entry point. Plural by signature.
└─ models.py	Django ORM only. No business logic, no cross-module ForeignKey.

- └─ registry.py
- └─ schedule.py
- └─ tasks.py

Module code, nav items and permission catalogue.
This module's beat entries, merged by config/celery.py.
Celery. Ids in arguments, never ORM objects.

Frontend — client/src/modules/placement/

└─ api/	The server contract, in one place.
└─ hooks.ts	TanStack Query. Every mutation invalidates what it can affect.
└─ types.ts	Wire types. Money and CPI stay STRINGS — Decimal, not float.
└─ components/	Reused within this module only. Anything shared goes to ui/.
└─ CpiBadge.tsx	A CPI never renders without its semester. null is not 0.00.
└─ DocumentsCard.tsx	Drive links. The raw URL never reaches the list.
└─ EligibilityPanel.tsx	Why a student may or may not apply, per rule.
└─ HistoryDrawer.tsx	An application's timeline. Says when reasons were withheld.
└─ IssueOfferModal.tsx	Issue an offer; the deadline is mandatory.
└─ NewPostingModal.tsx	The eligibility rule from named controls, not raw JSON.
└─ PlacementRecordCard.tsx	The rule 24 hold and the way to release it, side by side.
└─ pages/	One route each. Lazy-loaded, one chunk per page.
└─ AnnouncementsPage.tsx	Route: /placement/announcements
└─ ApplicationsPage.tsx	Route: /placement/applications
└─ CompaniesPage.tsx	Route: /placement/companies
└─ InterviewsPage.tsx	Route: /placement/interviews
└─ MyApplicationsPage.tsx	Route: /placement/mine
└─ OffersPage.tsx	Route: /placement/offers
└─ PostingsPage.tsx	Route: /placement/postings
└─ ProfilePage.tsx	Route: /placement/profile
└─ RegistrationApprovalsPage.tsx	Route: /placement/registration-approvals
└─ RegistrationPage.tsx	Route: /placement/registration
└─ ReportsPage.tsx	Route: /placement/reports
└─ StudentCpiPage.tsx	Route: /placement/students-cpi
└─ routes.tsx	Relative paths. Must match registry.py — CI compares them.

Shared scaffolding — use, do not fork

core/api/exceptions.py
core/api/csv.py
core/api/throttling.py
core/api/csrf.py
core/files/
fusion_auth/
client/src/ui/
client/src/app/registry.ts

The one error envelope. Views never build errors by hand.
Formula-injection-safe CSV for every module.
Principal-aware throttle.
CSRF for cookie-borne credentials.
Upload/download plumbing.
Identity. Modules read `request.principal`, never a user table.
Design system. FormModal, FormSection, DataTable, PageHeader.
Frontend module manifest. `code` must match the server.

Adding a module

1. Scaffold the directories below and add the app to `DOMAIN_MODULES`.
2. Write `registry.py`: module code, nav items, permission catalogue.
3. Model the domain in `domain/` first, in pure Python, with tests.
4. Add `models.py` and its migration. No cross-module FK.
5. Write selectors (reads) and services (writes). Services call `authz.require`.
6. Add the thin `api/` layer and mount it in `config/urls.py`.
7. Expose `contracts.py` — plural getters only.
8. Add the frontend manifest entry, routes and pages.
9. Run `make check`. It must be green before review.

A module granted to nobody is invisible and unroutable in production, so shipping half of one is safe. Grant it when it is ready.